

PROJECT SHOWCASE

Understanding Model Confidence & Loss

Using Entropy & Cross-Entropy in Fashion-MNIST

Information Theory | Deep Learning

Project Objective



Train Classifier: Build a simple MLP on Fashion-MNIST dataset.



Manual Computation: Implement Entropy & Cross-Entropy formulas.



Uncertainty Analysis: Quantify model confidence using probability.



Theoretical Bridge: Connect Info Theory to ML optimization.



Information & Surprise



Probability

Describes the raw uncertainty of outcomes in a system.



Surprise

Information measures how "surprising" an event is when it occurs.



Evaluation

ML uses these tools to evaluate how far predictions are from reality.

Shannon Information (Surprise)



Quantifying "Surprise"

$$I(x) = -\log(P(x))$$

- ✓ Rare events carry more information.
- ✓ High Probability = Low Surprise.
- ✓ Example: Winning the lottery has high information value.

Entropy: Measuring Uncertainty

Average Amount of Surprise

$$H(P) = - \sum_x p(x) \log p(x)$$

Entropy quantifies the uncertainty in a probability distribution.

[0.33, 0.33, 0.34] → High Entropy

[0.99, 0.005, 0.005] → Low Entropy

```
def entropy(probabilities):  
    epsilon = 1e-10  
  
    p = np.clip(  
        probabilities,  
        epsilon,  
        1.0  
    )  
  
    return -np.sum(  
        p * np.log(p)  
    )
```

Entropy Intuition

High Entropy

Model is Unsure: Probabilities are spread across multiple classes. High degree of disorder.

Visualization: Flattish distribution

Low Entropy

Model is Confident: One class clearly dominates.
Low degree of disorder.

Visualization: Spiky distribution

Important: Confidence does NOT mean correctness!

Cross-Entropy Loss

```
def custom_cross_entropy(y_true, y_pred):  
  
    epsilon = 1e-10  
  
    y_pred = tf.clip_by_value(  
        y_pred,  
        epsilon,  
        1.0  
    )  
  
    loss = -tf.reduce_sum(  
        y_true * tf.math.log(y_pred),  
        axis=1  
    )  
  
    return tf.reduce_mean(loss)
```

Compare Reality & Prediction

$$H(P, Q) = - \sum_x p(x) \log q(x)$$

P = True Distribution | Q = Predicted Distribution

Example: True = Cat, Pred = [0.8, 0.1, 0.1]

Loss = $-\log(0.8) \approx 0.097$

Why Use Cross-Entropy?



Rewards Accuracy

Rewards high probability assigned to the correct class.



Penalizes Errors

Heavily penalizes low probability on the correct class.



Useful Gradients

Provides strong signals for weight updates during backpropagation.

KL Divergence

P || Q
Distribution Distance

$$D_{\text{KL}} (P || Q) = \sum_x p (x) \log \left(\frac{p (x)}{q (x)} \right)$$

The Relationship

CE = Entropy + KL Divergence

$$H(P,Q) = H(P) + D_{\text{KL}}(P || Q)$$

Softmax & Prediction Flow



Experimental Setup

Category	Specification / Value
Dataset	Fashion-MNIST (60k Train / 10k Test)
Model Architecture	Multi-Layer Perceptron (MLP)
Hidden Layers	128 Neurons with ReLU Activation
Output Layer	10 Neurons with Softmax Activation
Hyperparameters	5 Epochs Adam Optimizer

Custom Implementations

```
def entropy(probabilities):  
    epsilon = 1e-10  
  
    p = np.clip(  
        probabilities,  
        epsilon,  
        1.0  
    )  
  
    return -np.sum(  
        p * np.log(p)  
    )
```

```
def cross_entropy_sample(  
    y_true,  
    y_pred):  
    epsilon = 1e-10  
  
    y_pred = np.clip(  
        y_pred,  
        epsilon,  
        1.0  
    )  
  
    return -np.sum(  
        y_true * np.log(y_pred)  
    )
```

Epsilon Shield: Added small constant to avoid $\log(0)$ errors.

Batched Compute: Processed 10,000 samples for entropy statistics.

Validation: Verified against built-in loss functions.

Model Performance



The simple MLP achieved solid classification performance with highly consistent Cross-Entropy metrics.

Entropy Results (Confidence)

Average Entropy by Correctness



Correct = High Confidence (Low Entropy)

Incorrect = Low Confidence (High Entropy)

Cross-Entropy Separation

0.096

CE Correct

2.210

CE Incorrect

Significant Separation

The cross-entropy loss reveals a massive gap between correctly and incorrectly classified items.

Low CE: Means the model assigned very high probability to the true class.

High CE: Means the model assigned very low probability to the actual target.

High vs Low Entropy Samples



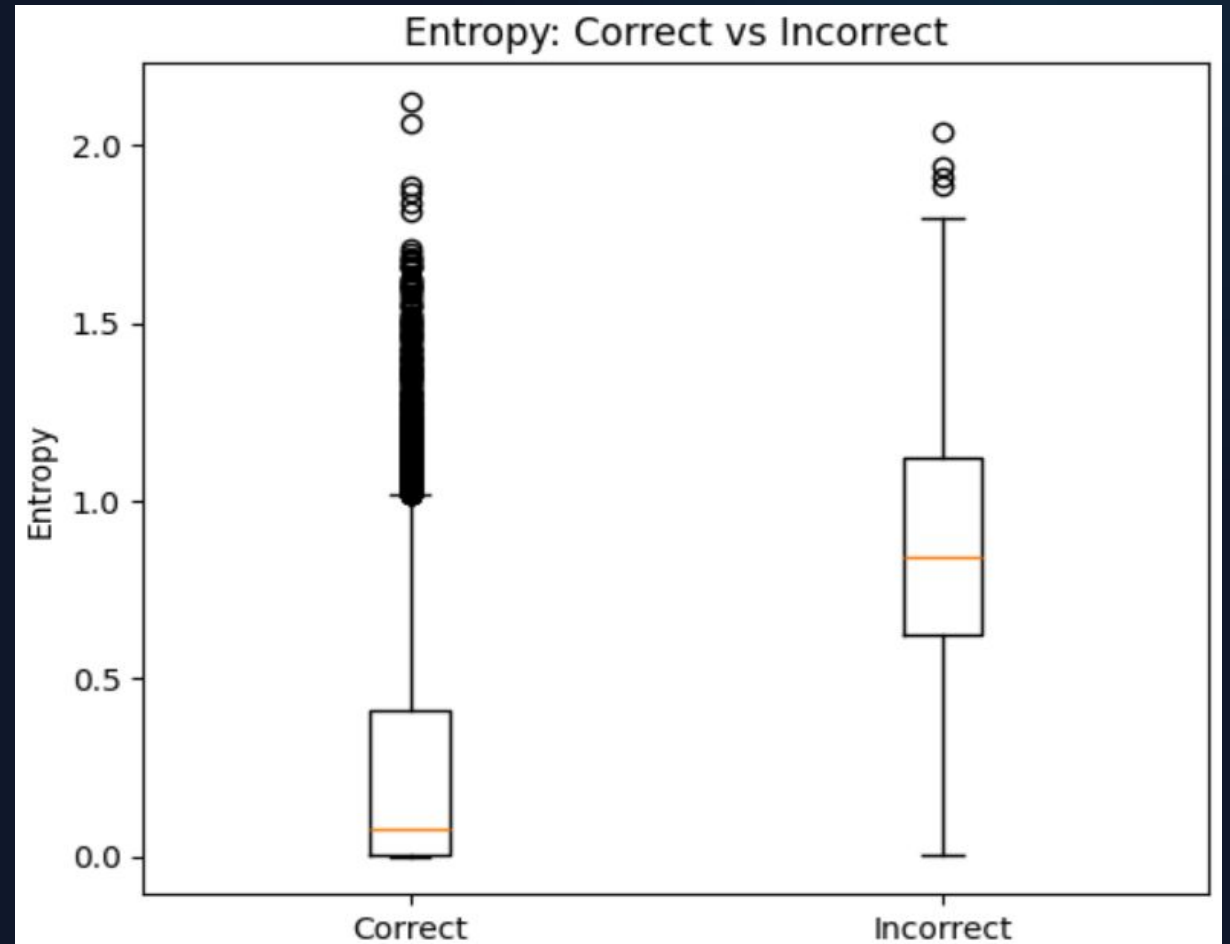
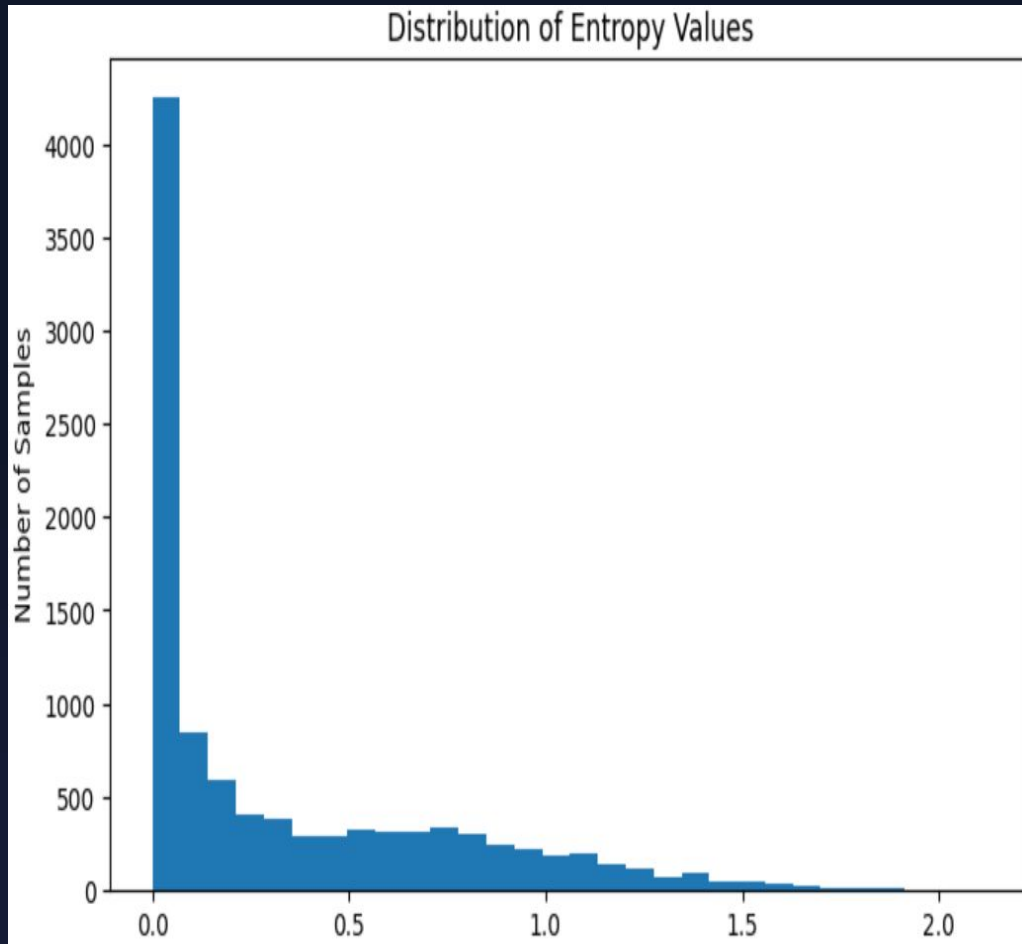
Low Entropy

Distinct items (like boots or bags) often yield confident, low-entropy predictions.

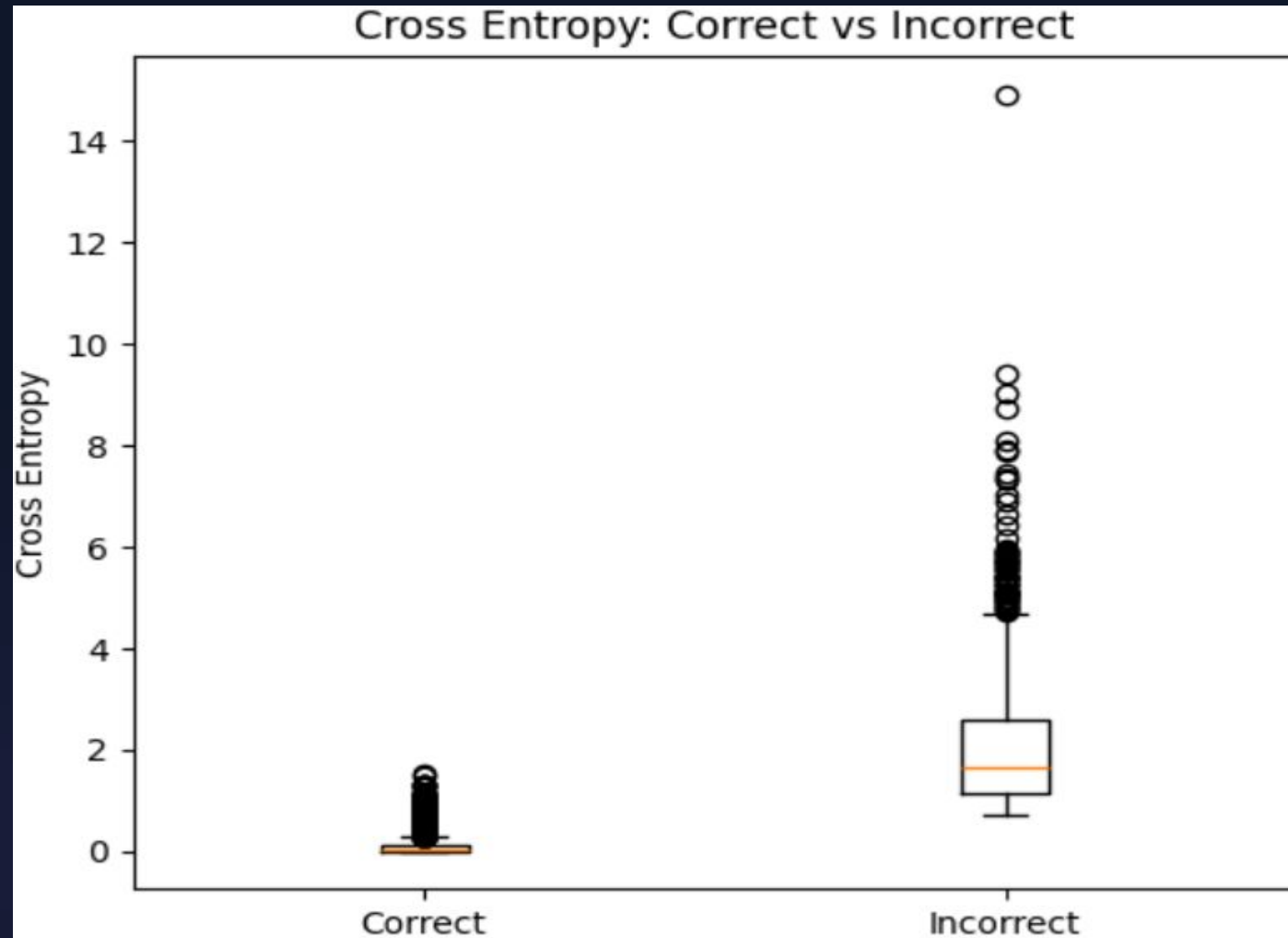
High Entropy

Overlapping classes (Pullover vs. Coat vs. Shirt) create confusion and flat probability spreads.

Visual Analysis



Visual Analysis

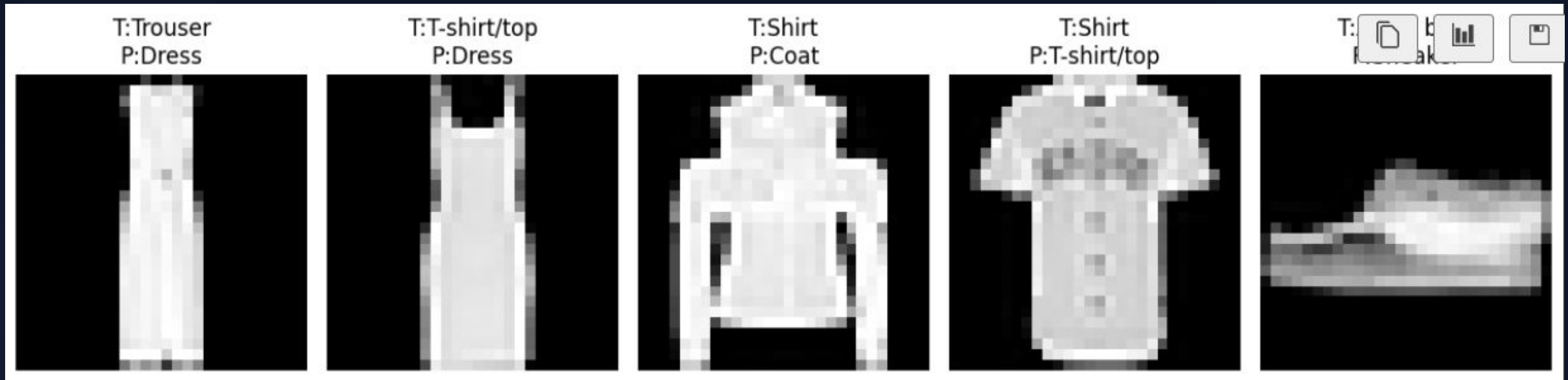


Critical Insight

"Low Entropy does not guarantee correctness. A model can be confidently, yet disastrously, wrong." —

Key Experimental Finding

— Key Experimental Finding



Key Learnings



Uncertainty

Entropy is the definitive metric for measuring raw uncertainty.



Distance

KL Divergence measures how different two worlds are.



Optimization

Cross-Entropy is the engine that drives model learning.

Challenges Faced

- ✓ **Mathematical Intuition:** Bridging logarithmic theory to actual pixel data.
- > **Custom Coding:** Ensuring numerical stability (avoiding log zero).
- 👁️ **Interpretation:** Understanding why specific clothing items confused the model.
- ⚖️ **Confidence Bias:** Recognizing that confidence $\neq$ correctness.

